A Course End Project

LED Dimmer using PWM

IOT LABORATORY (A8603)

BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING

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This is to certify that the Course End Project titled **"Led dimmer using PWM"** is carried out by **K. Anusha (24881A05A3), T. Harshini (24881A05C2)**, towards **A8603 – IOT Laboratory** course in partial fulfilment of the requirements for the award of degree of **Bachelor of Technology in Computer Science and Engineering** during the Academic Year 2025–26.

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Check List

S. No.	Content	Status (✓ / ✗)
1	Abstract	✓
2	Objectives	✓
3	Problem Statement	✓
4	Algorithm	✓
5	Flow Chart	✓
6	Source Code	✓
7	Input	✓
8	Output	✓
9	Conclusion	✓
10	Future Scope	✓
11	References	✓

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1 Abstract

This project titled *"LED Dimmer using PWM"* focuses on designing an efficient and flexible system for controlling the brightness of an LED using Pulse Width Modulation (PWM) technique. With the increasing demand for energy-efficient lighting systems and smart control mechanisms, this project demonstrates how digital control methods can be used to regulate power without energy loss.

The proposed system utilizes an Arduino microcontroller to generate PWM signals, which control the intensity of the LED. A potentiometer is used as an input device to vary the brightness level manually. The Arduino reads the analog input from the potentiometer, processes the signal using its built-in Analog-to-Digital Converter (ADC), and converts it into a PWM output signal.

Unlike traditional dimming techniques that rely on resistive methods and result in energy loss as heat, PWM ensures efficient power utilization by rapidly switching the signal ON and OFF. This switching occurs at a frequency high enough that the human eye perceives it as continuous brightness variation.

The system is simple, cost-effective, and highly efficient, making it suitable for applications such as smart lighting systems, industrial controls, and home automation. From an embedded systems perspective, the project demonstrates practical implementation of PWM, analog signal processing, and real-time control using microcontrollers.

Keywords: PWM, Arduino, LED Dimming, Duty Cycle, Embedded Systems, Analog-to-Digital Conversion, Smart Lighting, Power Efficiency.

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2 Problem Statement

In conventional lighting systems, brightness control is typically achieved using resistors or variable voltage regulators. These methods, although simple, suffer from several significant limitations. One of the major drawbacks is the loss of energy in the form of heat, which reduces overall efficiency. Additionally, such systems provide limited control over brightness and may affect the lifespan of LEDs due to inconsistent power supply.
As modern technology advances toward energy-efficient and smart systems, there is a growing need for better methods of controlling electrical devices. Traditional dimming methods are not suitable for applications requiring precise, smooth, and efficient brightness control. They also lack flexibility and adaptability in automated environments.

Therefore, there is a need for a system that can:

- Provide efficient power control without energy loss
- Ensure smooth and precise brightness variation
- Offer user-friendly operation
- Be easily integrated into embedded and IoT systems

The LED Dimmer using PWM addresses these challenges by using digital control techniques, ensuring high efficiency, reliability, and flexibility.

3 Objectives

- To understand the concept and working of Pulse Width Modulation (PWM).
- To design a system for controlling LED brightness using PWM signals.
- To interface a potentiometer with Arduino for user input.
- To implement analog-to-digital conversion using Arduino.
- To achieve smooth and efficient brightness control.
- To reduce power loss compared to traditional dimming methods.
- To demonstrate practical application of embedded systems and microcontroller programming

4 Algorithm

1. **System Initialization:**
   a) Power ON the system
   b) Initialize Arduino microcontroller
   c) Configure LED pin as OUTPUT
2. **Read Input:**
   a) Continuously read analog value from potentiometer
   b) Convert voltage into digital value (0–1023)
3. **Processing:**
   a) Map analog value to PWM range (0–255)
   b) Determine duty cycle
4. **Control Action:**
   a) Generate PWM signal
   b) Send PWM signal to LED
5. **Output:**
   a) Adjust LED brightness based on duty cycle
6. **Continuous Operation:**
   a) Repeat the process continuously
   b) Provide real-time brightness control
7. **End:**
   a) System runs continuously until powered OFF

5 Proposed System

The proposed system implements a PWM-based LED dimming mechanism using an Arduino microcontroller and a potentiometer. The system is designed to provide efficient and smooth brightness control by varying the duty cycle of the PWM signal.

The potentiometer acts as a user input device, allowing manual adjustment of brightness. The Arduino reads the analog signal from the potentiometer and converts it into a digital value. This value is then used to generate a PWM signal, which controls the LED intensity.

The system operates continuously and responds instantly to changes in input. Since PWM does not waste energy as heat, the system is highly efficient and reliable. It also ensures longer lifespan of components compared to traditional dimming methods.

The design is simple and cost-effective, making it suitable for small-scale applications and educational purposes. It can also be extended to advanced systems such as IoT-based smart lighting and automation.

5.1 Flow Chart

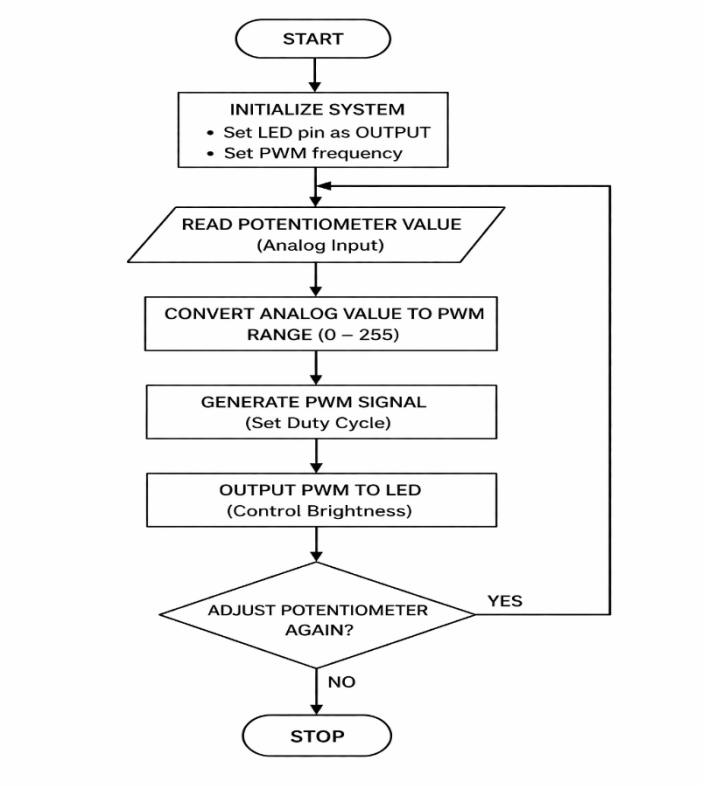


Figure 1: Flow Chart

5.2 Working Model

• **Objective:** To control the brightness of an LED using Pulse Width Modulation (PWM) through an Arduino microcontroller, providing an efficient and smooth alternative to traditional voltage-based dimming methods.

• **Components Required:**
– Microcontroller: Arduino UNO
– Input Device:
∗ Potentiometer (10kΩ)
– Output Device:
∗ LED (for brightness indication)
– Resistor (220Ω for LED protection)
– Power Supply (5V via USB)
– Connecting wires and breadboard

• **System Workflow:**
– Initialization: Arduino powers ON and initializes LED pin and analog input pin.
– Data Monitoring: Arduino continuously reads analog values from the potentiometer.

– Input Processing**:**
  ∗ Potentiometer provides variable voltage (0–5V).
  ∗ Arduino converts it into digital values (0–1023).

– **PWM Conversion:**
  ∗ Analog value is mapped to PWM range (0–255).

– **Control Action:**
  ∗ PWM signal is generated based on mapped value.
  ∗ LED brightness increases or decreases accordingly.

– **Continuous Operation:** System continuously reads input and updates LED brightness in real-time.

• **Implementation Steps:**
– Connect potentiometer (VCC, GND, middle pin) to Arduino (5V, GND, A0).
– Connect LED with resistor to Arduino PWM pin (Pin 9).
– Write and upload Arduino code to read analog input and generate PWM output.
– Rotate potentiometer knob to vary input voltage.
– Observe LED brightness change and verify smooth operation.

• **Advantages of This Working Model:**
– Provides smooth and precise brightness control
– Reduces power loss compared to traditional methods
– Energy efficient and reliable
– Simple and cost-effective design
– Easy to implement and maintain
– Can be extended for IoT-based smart lighting systems

6 Source Code

```
#define LED_PIN 9        // PWM pin connected to LED
#define POT_PIN A0       // Analog pin connected to potentiometer
int potValue = 0;        // Variable to store potentiometer value
int brightness = 0;      // Variable to store PWM value

void setup() {
  Serial.begin(9600);   // Initialize serial communication (optional)

  pinMode(LED_PIN, OUTPUT);   // Set LED pin as output

  Serial.println("LED Dimmer using PWM Ready");
}

void loop() {

  // Read analog value from potentiometer (0–1023)
  potValue = analogRead(POT_PIN);

  // Convert analog value to PWM range (0–255)
  brightness = map(potValue, 0, 1023, 0, 255);

  // Generate PWM signal to control LED brightness
  analogWrite(LED_PIN, brightness);

  Serial.print("Pot Value: ");
  Serial.print(potValue);
  Serial.print("  Brightness: ");
  Serial.println(brightness);

  delay(10); // Small delay for smooth operation
}
```

7 Input

The LED Dimmer using PWM system takes the following inputs to operate efficiently:

• **Potentiometer Input:** The potentiometer acts as the primary input device and provides a variable analog voltage based on the position of its knob. This voltage determines the brightness level of the LED.

• **Analog Signal:** The potentiometer generates an analog signal in the range of 0V to 5V. This signal is continuously read by the Arduino through its analog input pin and converted into a digital value (0–1023) using the built-in ADC.

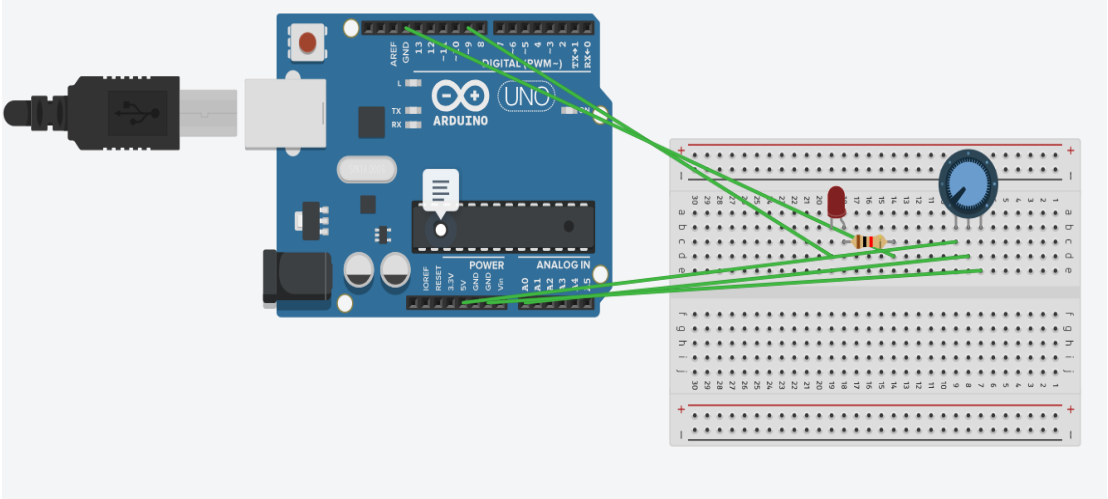
• **Power Supply:** A 5V power supply from the Arduino is required to operate the entire system, including the potentiometer and LED circuit.

• **User Interaction:** The system relies on manual user input through the rotation of the potentiometer knob, allowing smooth and real-time control of LED brightness.

• **Timing Control (Optional):** A small delay in the program ensures stable readings from the potentiometer and provides smooth transitions in LED brightness without flickering.

These inputs enable the system to accurately read user-controlled analog values, process them through the Arduino, and generate a corresponding PWM signal to control LED brightness efficiently and reliably.

8 Output



The output of the system demonstrates a simple and effective PWM-based brightness control mechanism, replacing traditional voltage-based dimming methods with a modern, efficient, and responsive electronic solution.

The LED brightness varies smoothly based on the input from the potentiometer. As the potentiometer is adjusted, the Arduino generates a corresponding PWM signal, which controls the intensity of the LED in real-time.

- At minimum input → LED remains OFF or very dim
- At medium input → LED glows with moderate brightness
- At maximum input → LED glows at full brightness

The rapid switching of the PWM signal is not visible to the human eye, resulting in a smooth and continuous variation in brightness.

This output confirms the successful implementation of Pulse Width Modulation for efficient power control, ensuring low energy consumption, reduced heat loss, and reliable performance.

9 Conclusion

The LED Dimmer using PWM successfully demonstrates an efficient and modern method for controlling LED brightness using digital techniques. By utilizing Pulse Width Modulation, the system is able to regulate the power delivered to the LED without significant energy loss, overcoming the limitations of traditional voltage-based dimming methods.

The integration of a potentiometer with the Arduino microcontroller allows real-time user control over brightness, making the system simple, interactive, and user-friendly. The smooth variation in LED intensity confirms the effectiveness of PWM in providing precise and continuous control.

One of the major advantages of this system is its high efficiency, as it minimizes power dissipation in the form of heat and improves the lifespan of components. The design is cost-effective, easy to implement, and requires minimal hardware, making it suitable for both educational purposes and practical applications.

From an embedded systems perspective, the project provides a clear understanding of analog-to-digital conversion, PWM signal generation, and hardware-software integration. It also highlights the importance of efficient power management in modern electronic systems.

Overall, the LED Dimmer using PWM is a reliable and scalable solution that forms a strong foundation for advanced applications such as smart lighting systems, home automation, and IoT-based control systems.

10 Future Scope

The LED Dimmer using PWM project can be further enhanced and extended to support more advanced and intelligent applications in the field of embedded systems and IoT.

• The system can be integrated with **IoT platforms** to enable remote control of LED brightness using mobile applications or web interfaces.

• Wireless communication technologies such as **Bluetooth** or **Wi-Fi** can be incorporated for convenient and flexible operation.

• The project can be expanded to control **multiple LEDs or lighting systems**, making it suitable for home and industrial lighting automation.

• Advanced sensors such as **light sensors (LDR)** can be added to implement **automatic brightness control** based on ambient light conditions.

• Voice control features can be integrated using virtual assistants like **Google Assistant** or **Alexa** for hands-free operation.

• The system can be adapted for **energy-efficient smart lighting systems**, contributing to reduced power consumption and sustainable solutions.

• Integration with **mobile apps** can provide features like scheduling, brightness presets, and usage monitoring.

• The design can be implemented using more advanced microcontrollers or platforms such as **ESP8266/ESP32** for enhanced performance and connectivity.

11 References

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